Resources ▼

format

Set output display format

Syntax

```
format(style)

fmt = format
fmt = format(style)
```

Description

`format(style)` changes the output display format to the format specified by `style`. For example, `format("shortG")` displays numeric values in a compact form with 5 total digits. Numeric formats affect only how numbers appear in the display, not how MATLAB® computes or saves them. example

When you specify the style by name, you can use *command form* without parentheses or quotes:

```
format shortG
```

You can set the format in either the Command Window or the Live Editor. Changing the format in one automatically updates it in both contexts.

`fmt = format` returns the current display format. (*since R2021a*) example

`fmt = format(style)` stores the current display format in `fmt` and then changes the display format to the specified style. (*since R2021a*) example

You cannot use command form when you request output or when you pass a variable as input. Enclose inputs in parentheses and include style names in quotes.

```
fmt = format("shortG");
format(fmt)
```

Examples

collapse all

Long Format

Set the output format to the long fixed-decimal format and display the value of `pi`.

Open in MATLAB Online

Copy Command

```
format long
pi
```

Get

```
ans =
   3.141592653589793
```

Hexadecimal Format

Display the maximum values for integers and real numbers in hexadecimal format.

Open in MATLAB Online

Copy Command

```
format hex
intmax('uint64')
```

Get

```
ans = uint64
   ffffffffffffffff
```

```
realmax
```

Get

```
ans =
   7fefffffffffffff
```

Short and Long Engineering Notation

Display the difference between `shortEng` and `longEng` formats.

Set the output format to `shortEng`.

Open in MATLAB Online

Copy Command

```
format shortEng
```

Get

Create a variable and increase its value by a multiple of 10 each time through a `for` loop.

```
A = 5.123456789;
for k = 1:10
   disp(A)
   A = A*10;
end
```

Get

```
     5.1235e+000

    51.2346e+000

   512.3457e+000

     5.1235e+003
```

```
     51.2346e+003

    512.3457e+003

      5.1235e+006

     51.2346e+006

    512.3457e+006

      5.1235e+009
```

The values display with 4 digits after the decimal point and an exponent that is a multiple of 3.

Set the output format to the long engineering format and view the same values.

```
format longEng

A = 5.123456789;
for k = 1:10
   disp(A)
   A = A*10;
end
```

Get

```
     5.12345678900000e+000

     51.2345678900000e+000

     512.345678900000e+000

     5.12345678900000e+003

     51.2345678900000e+003

     512.345678900000e+003

     5.12345678900000e+006

     51.2345678900000e+006

     512.345678900000e+006

     5.12345678900000e+009
```

The values display with 15 digits and an exponent that is a multiple of 3.

Large Data Range Format

Use the `shortG` format when some of the values in an array are short numbers and some have large exponents. The `shortG` format picks whichever short fixed-decimal format or short scientific notation has the most compact display.

Open in MATLAB Online

Create a variable and display output in the `short` format, which is the default.

Copy Command

```
x = [25 56.31156 255.52675 9876899999];
format short
x
```

Get

```
x = 1×4
10^9 ×

    0.0000    0.0000    0.0000    9.8769
```

Set the format to `shortG` and redisplay the values.

```
format shortG
x
```

Get

```
x = 1×4

            25        56.312        255.53    9.8769e+09
```

Reset Format to Default

Set the output format to the short engineering format with compact line spacing.

```
format shortEng
format compact
x = rand(3)
```

```
x =
   814.7237e-003   913.3759e-003   278.4982e-003
   905.7919e-003   632.3592e-003   546.8815e-003
   126.9868e-003    97.5404e-003   957.5068e-003
```

Reset the display format to default and display the matrix again.

```
format default
x
```

```
x =

    0.8147    0.9134    0.2785
    0.9058    0.6324    0.5469
    0.1270    0.0975    0.9575
```

Before R2021a, reset the display format to default values using `format` by itself.

```
format
```

Get Current Format

Since R2021a

Get the current display format.

Open in MATLAB Online

Copy Command

```
fmt = format
```

Get

```
fmt = 
  DisplayFormatOptions with properties:

    NumericFormat: "short"
      LineSpacing: "loose"
```

Save and Restore Display Format

Since R2021a

Save the current display format and restore it at a later time.

Set the numeric display to `shortE` and display a 2-by-2 matrix of numeric values.

Open in MATLAB Online

Copy Command

```
format shortE
m = [9638573934 37467; 236 574638295]
```

Get

```
m = 2×2

   9.6386e+09   3.7467e+04
   2.3600e+02   5.7464e+08
```

Save the current display format in `oldFmt` and change the numeric format to `longE`.

```
oldFmt = format("longE")
```

Get

```
oldFmt = 
  DisplayFormatOptions with properties:

    NumericFormat: "shortE"
      LineSpacing: "loose"
```

Confirm that the numeric format is now long, scientific notation by redisplaying matrix `m`.

```
m
```

Get

```
m = 2×2

     9.638573934000000e+09     3.746700000000000e+04
     2.360000000000000e+02     5.746382950000000e+08
```

Restore the format to its previous state. Redisplay `m` to confirm that the numeric format is now short, scientific format.

```
format(oldFmt)
m
```

Get

```
m = 2×2

   9.6386e+09   3.7467e+04
   2.3600e+02   5.7464e+08
```

Input Arguments

collapse all

`style` — Format to apply
character vector | string scalar | `DisplayFormatOptions` object

Format to apply, specified as a character vector, string scalar, or `DisplayFormatOptions` object.

Character vectors or string scalars must be one of the listed style names or `default`.

Default

`default` restores the default display format, which is `short` for numeric format and `loose` for line spacing. (*since R2021a*)

Numeric Format

These styles control the output display format for numeric variables.

`Style`	Result	Example
`short`	Short, fixed-decimal format with 4 digits after the decimal point. This is the default numeric setting.	`3.1416`
`long`	Long, fixed-decimal format with 15 digits after the decimal point for `double` values, and 7 digits after the decimal point for `single` values.	`3.141592653589793`
`shortE`	Short scientific notation with 4 digits after the decimal point.	`3.1416e+00`
`longE`	Long scientific notation with 15 digits after the decimal point for `double` values, and 7 digits after the decimal point for `single` values.	`3.141592653589793e+00`
`shortG`	Short, fixed-decimal format or scientific notation, whichever is more compact, with a total of 5 digits.	`3.1416`
`longG`	Long, fixed-decimal format or scientific notation, whichever is more compact, with a total of 15 digits for `double` values, and 7 digits for `single` values.	`3.14159265358979`
`shortEng`	Short engineering notation (exponent is a multiple of 3) with 4 digits after the decimal point.	`3.1416e+000`

Style	Result	Example
`longEng`	Long engineering notation (exponent is a multiple of 3) with 15 significant digits.	`3.14159265358979e+000`
`+`	Positive/Negative format with `+`, `-`, and blank characters displayed for positive, negative, and zero elements.	`+`
`bank`	Currency format with 2 digits after the decimal point.	`3.14`
`hex`	Hexadecimal representation of a binary double-precision number.	`400921fb54442d18`
`rational`	Ratio of small integers.	`355/113`

Line Spacing Format

Style	Result	Example

`compact`	Suppress excess blank lines to show more output on a single screen.	`theta = pi/2`
		`theta =`
		`  1.5708`

`loose`	Add blank lines to make output more readable. This is the default setting for line spacing.	`theta = pi/2`
		
		`theta =`
		
		`  1.5708`

The `DisplayFormatOptions` object has two properties, `NumericFormat` and `LineSpacing`. The options for character vector and string scalar inputs are also the valid property values. For an example of using a `DisplayFormatOptions` object, see Save and Restore Display Format.

Output Arguments

collapse all

`fmt` — Current display format
`DisplayFormatOptions` object

Current display format, returned as a `DisplayFormatOptions` object with these properties:

- `NumericFormat`
- `LineSpacing`

For valid property values, see the `style` argument.

> **Note**
>
> Property values reflect the state of the display format when the object is created. The properties do not automatically change when the display format changes. See Save and Restore Display Format for an example.

Tips

- The specified format applies only to the current MATLAB session. To maintain a format across sessions, choose a **Numeric format** or **Line spacing** option in the Command Window settings.
- You can specify `short` or `long` and the presentation type separately, such as `format short E` or `format("short E")`.
- MATLAB always displays integer data types to the appropriate number of digits for the data type. For example, MATLAB uses 3 digits to display `int8` data types (for instance, -128:127). Setting the output format to `short` or `long` does not affect the display of integer-type variables.
- Integer-valued, floating-point numbers with a maximum of 9 digits do not display in scientific notation.
- If you are displaying a matrix with a wide range of values, consider using `shortG`. See Large Data Range Format.

Extended Capabilities

expand all

Thread-Based Environment
Run code in the background using MATLAB® `backgroundPool` or accelerate code with Parallel Computing Toolbox™ `ThreadPool`.

Version History

expand all

Introduced before R2006a

R2021a: `format` with no arguments is not recommended

See Also

`DisplayFormatOptions` | `disp` | `fprintf` | `formattedDisplayText`

Topics

Format Output

Modify Workspace and Variables Settings